Barenblatt Similarity Profiles for Every Positive Exponent: Exact Mass, Moment, Free-Boundary, and Dissipation Atlas

Route-A structural certificate C207

Abstract
For the one-dimensional equation $u_t = (u^m)_{xx}$, every $m > 0$ and mass $M > 0$, we classify the centered nonnegative integrable zero-flux first-kind similarity profiles. One derivation yields compact support for $m > 1$, the Gaussian at $m = 1$, and algebraic tails for $0 < m < 1$. Exact Beta integrals normalize mass and give every absolute moment, including the sharp threshold $r < (1 + m)/(1 - m)$ and the logarithmic second-moment boundary $m = 1/3$. We also give porous pressure/free-boundary data and a conditional rescaled dissipation identity. The statement does not classify arbitrary Cauchy solutions. Exact finite cells audit conventions but do not prove the continuous theorem. Every Route-A gate fails.

Keywords: porous-medium equation; fast diffusion; Barenblatt profile; similarity solution; moment threshold.

中文摘要

本文对一维方程$u_t = (u^m)_{xx}$的全部$m > 0$和给定质量，分类中心化、非负、可积、零通量的第一类自相似剖面。统一推导覆盖紧支撑、高斯与代数尾三种区域，并给出精确质量、全阶绝对矩、$m = 1/3$二阶矩边界、压力自由边界及限定适用范围的重标度耗散。结论不扩张为任意初值问题解的分类；路线A全部门失败。

1 Frozen class and profile equation

Let $m > 0$, $M > 0$, and consider

$$u_t = (u^m)_{xx}, \qquad x \in \mathbb{R}, \quad t > 0. \tag{1}$$

We classify only centered, nonnegative, integrable first-kind profiles of mass M such that $F^m \in W^{1,1}_{\mathrm{loc}}(\mathbb{R})$ and the integrated zero-flux law below holds almost everywhere. Uniqueness is up to almost-everywhere equality (equivalently, for the continuous representative determined by F^m). Translates, signed profiles, higher dimensions, and arbitrary Cauchy solutions are outside the claim. Source-type ownership and the surrounding theory are classical [1, 2]; no formula priority is claimed.

Mass-preserving scaling has

$$u(x, t) = t^{-\alpha} F(\xi), \qquad \xi = xt^{-\alpha}, \qquad \alpha = \frac{1}{m+1}.$$

Indeed exponent matching gives $\alpha + 1 = m\alpha + 2\alpha$. The profile equation (in distributions) and its zero-flux integral (almost everywhere) are

$$-\alpha(F + \xi F') = (F^m)'', \qquad (F^m)' + \alpha\xi F = 0. \tag{2}$$

2 All positive exponents and exact mass

Theorem 1. *In the frozen class there is exactly one mass-M profile. For $m > 1$, set*

$$p = \frac{1}{m-1}, \qquad k = \frac{m-1}{2m(m+1)}.$$

Then

$$F(\xi) = (C - k\xi^2)_+^p, \qquad M = C^{p+1/2} k^{-1/2} \mathrm{B}\left(\frac{1}{2}, p+1\right). \tag{3}$$

For $m = 1$,

$$F(\xi) = \frac{M}{2\sqrt{\pi}} \mathrm{e}^{-\xi^2/4}. \tag{4}$$

For $0 < m < 1$, set

$$q = \frac{1}{1-m}, \qquad b = \frac{1-m}{2m(m+1)}.$$

Then

$$F(\xi) = (C + b\xi^2)^{-q}, \qquad M = C^{1/2-q} b^{-1/2} \mathrm{B}\left(\frac{1}{2}, q - \frac{1}{2}\right). \tag{5}$$

In each nonlinear regime the displayed equation uniquely fixes $C > 0$.

Proof. Where $F > 0$, equation (2) gives

$$(F^{m-1})' = -\frac{\alpha(m-1)}{m}\xi \quad (m \neq 1), \qquad (\log F)' = -\alpha\xi \quad (m = 1).$$

Integration on each positivity component gives (3)–(5). Local absolute continuity makes F continuous. In the porous case a finite endpoint forces $F^{m-1} = C - k\xi^2 = 0$; the two endpoints therefore have equal modulus, so the only nonempty component is $(-\sqrt{C/k}, \sqrt{C/k})$. In the fast and heat cases the displayed finite expressions cannot approach zero at a finite component endpoint, while integrability excludes a missing infinite end; their positivity set is all of $\mathbb{R}$. Thus no extra positivity component or zero interval is possible. Substituting $z = k\xi^2/C$ or $z = b\xi^2/C$ evaluates the two mass integrals by Euler's Beta integral; Gaussian integration gives (4). The porous mass is strictly increasing in C, while the fast mass is strictly decreasing because $q > 1$. Thus mass fixes C. Conversely the formulas satisfy (2), have mass M, and are centered. This proves existence and uniqueness inside the frozen profile class. □

3 All absolute moments and the sharp tail boundary

Proposition 1. *For every $r > -1$, the porous and Gaussian absolute moments are respectively*

$$\int_{\mathbb{R}} |\xi|^r F_{\mathrm{por}}(\xi)\, d\xi = C^{p+(r+1)/2} k^{-(r+1)/2} \mathrm{B}\left(\frac{r+1}{2}, p+1\right). \tag{6}$$

For the Gaussian,

$$\int_{\mathbb{R}} |\xi|^r F_{\mathrm{G}}(\xi)\, d\xi = M\, 2^r \frac{\Gamma((r+1)/2)}{\sqrt{\pi}}. \tag{7}$$

Both are finite. In fast diffusion the moment is finite exactly when

$$r < 2q - 1 = \frac{1+m}{1-m}, \tag{8}$$

and then equals

$$\int_{\mathbb{R}} |\xi|^r F_{\mathrm{fast}}(\xi)\, d\xi = C^{-q+(r+1)/2} b^{-(r+1)/2} \mathrm{B}\left(\frac{r+1}{2}, q - \frac{r+1}{2}\right). \tag{9}$$

At equality divergence is logarithmic; above it divergence is a power. Hence the fast second moment is finite exactly for $m > 1/3$, logarithmically divergent at $m = 1/3$.

Proof. Evenness and the substitutions used for mass give (6)–(9). Alternatively, $F(\xi) \asymp |\xi|^{-2q}$ in the fast regime, so the tail integral has power $r - 2q$ and converges exactly when $r - 2q < -1$. Equality yields $\int^R d\xi/\xi$; the second-moment statement follows by solving $2 < (1+m)/(1-m)$. ☐

4 Pressure, free boundary, and rescaled dissipation

For $m > 1$, the pressure

$$P = \frac{m}{m-1} u^{m-1}$$

is quadratic on the positivity set. If $R_M = \sqrt{C/k}$, then $\operatorname{supp} u(\cdot, t) = [-R_M t^\alpha, R_M t^\alpha]$ and the two interfaces $X_\pm(t) = \pm R_M t^\alpha$ obey the exact one-sided pressure law

$$X'_\pm(t) = \frac{\alpha X_\pm(t)}{t} = -\lim_{x \to X_\pm(t),\, u(x,t)>0} P_x(x,t).$$

Put $\tau = \log t$, $\xi = x t^{-\alpha}$, and $v = t^\alpha u$. Then

$$v_\tau = (v^m)_{\xi\xi} + \alpha(\xi v)_\xi = \partial_\xi(v \partial_\xi \mu), \tag{10}$$

where

$$\mu = \frac{m}{m-1} v^{m-1} + \frac{\alpha}{2}\xi^2 \ (m \neq 1), \qquad \mu = \log v + \frac{\alpha}{2}\xi^2 \ (m = 1).$$

This is the first variation of the explicitly branched free energy

$$\mathcal{F}_m[v] = \begin{cases} \displaystyle\int_{\mathbb{R}} \left(\frac{v^m}{m-1} + \frac{\alpha}{2}\xi^2 v\right) d\xi, & m \neq 1, \\ \displaystyle\int_{\mathbb{R}} \left(v \log v - v + \frac{1}{4}\xi^2 v\right) d\xi, & m = 1. \end{cases} \tag{11}$$

Every profile in Theorem 1 is stationary. For sufficiently regular positive rescaled solutions for which the displayed terms in (11) are finite (so no $\infty - \infty$ is used) and whose decay justifies integration by parts,

$$\frac{d\mathcal{F}_m}{d\tau} = \int_{\mathbb{R}} \mu\, \partial_\xi(v \partial_\xi \mu)\, d\xi = -\int_{\mathbb{R}} v |\partial_\xi \mu|^2\, d\xi. \tag{12}$$

This identity is not asserted outside that class. For the fast Barenblatt profiles, the finite-second-moment/free-energy setting used here starts at $m > 1/3$. At $m \leq 1/3$ that functional is not finite on the Barenblatt profile, so equation (12) is not applied to it.

Revision focus. Round 1 adds every absolute moment, the sharp tail and $m = 1/3$ boundaries, pressure/free-boundary data, and conditional rescaled dissipation from the explicitly branched free energy.

Declarations

Data and code availability. Package-local exact synthetic evidence and deterministic code accompany the paper; no observational data are used.

Ethics. No human, animal, personal, or sensitive data are used.

Author contributions (CRediT). This anonymous certificate records Conceptualization, Formal analysis, Software, Validation, Writing—original draft, and Writing—review and editing. AI systems are not authors.

Funding. No external funding is reported.

Conflicts of interest. None known.

AI-use disclosure. An AI coding assistant supported derivation, drafting, and exact-code development; it was not an external reviewer or an independent peer-review process.

References

[1] G. I. Barenblatt, "On some unsteady motions of a liquid and gas in a porous medium," *Prikl. Mat. Mekh.* 16 (1952), 67–78.

[2] J. L. Vázquez, *The Porous Medium Equation: Mathematical Theory*, Oxford University Press, 2007. DOI: `10.1093/acprof:oso/9780198569039.001.0001`.